A Practical Training Report on

# CYBER THREAT AND PHISHING DETECTION USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING

Submitted in partial fulfilment of the requirements for the award of the degree of

**Bachelor of Technology**
**in**
**Electronics & Computer Science Engineering**

by

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Under the guidance of

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**SCHOOL OF ELECTRONICS ENGINEERING**

**KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY**

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# CERTIFICATE OF PRACTICAL TRAINING

The Practical Training Completion Certificate was issued by the School of Computer Applications, Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, Bhubaneswar – 751024, Odisha, certifying completion of the 8-Week Summer Internship in AI/ML, Cybersecurity, Image Processing, and NLP from 04.05.2026 to 04.07.2026.

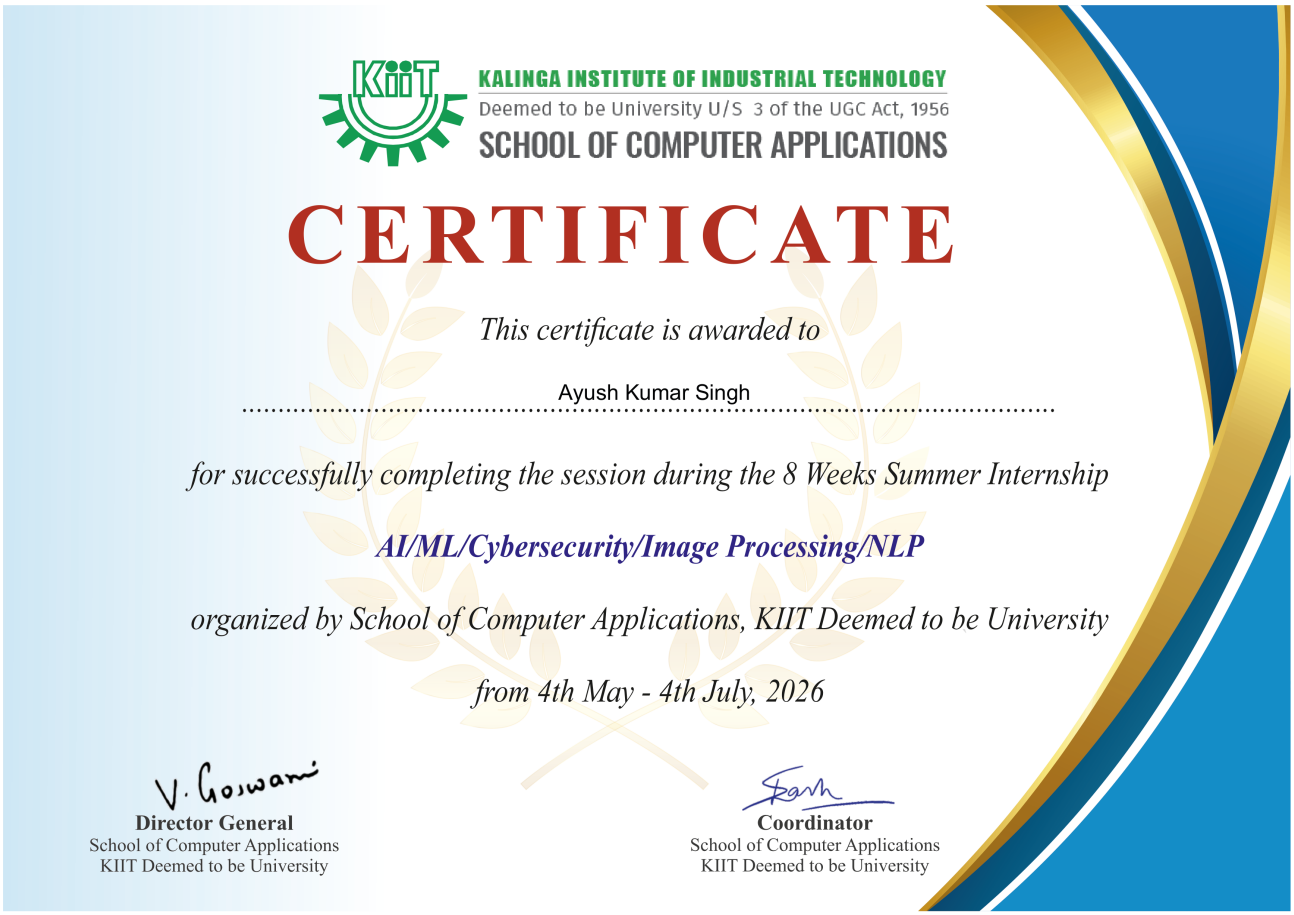



*Figure 0.1: Official Practical Training Completion Certificate received from School of Computer Applications, KIIT Deemed to be University (dated 04/07/2026).*

*Copy of the original certificate — Candidate: Ayush Kumar Singh (University Roll Number: 2330372, B.Tech E&CSE;, KIIT-DU).*

# CANDIDATE'S DECLARATION

I, Ayush Kumar Singh (University Roll Number: 2330372), a bonafide undergraduate student of the Bachelor of Technology (B.Tech) programme in Electronics & Computer Science Engineering at the School of Electronics Engineering, Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, Bhubaneswar – 751024, Odisha, hereby declare that the Practical Training Report entitled "Cyber Threat and Phishing Detection using Natural Language Processing and Machine Learning" is an authentic and original record of the practical training work carried out by me during the 8-Week Summer Internship (04 May 2026 to 04 July 2026) organized by the School of Computer Applications, KIIT Deemed to be University.

I further declare that the research and development of the natural language processing pipeline, sublinear TF-IDF n-gram feature extraction, Multinomial Naive Bayes probabilistic baseline modeling, Random Forest ensemble classification, Explainable AI (XAI) feature keyword attribution, and Streamlit security dashboard operations, as presented in this report, were carried out by me under the supervision of Prof. (Dr.) Satya Ranjan Dash and comply with the academic regulations of KIIT Deemed to be University.

The contents of this report have not been submitted, either in part or in full, to any other university or educational institution for the award of any degree, diploma, certificate or fellowship. All external technical standards, machine learning frameworks, and cybersecurity references consulted during the preparation of this work have been duly acknowledged in the References section.

## SUMMARY OF CANDIDATE & SUBMISSION DETAILS

| | |
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| **Training Organisation & Unit** | School of Computer Applications, KIIT Deemed to be University (AI/ML/Cybersecurity/NLP) |
| **Training Period** | 04.05.2026 to 04.07.2026 (8 weeks) |

Place: Bhubaneswar, Odisha
Date: 04 July 2026

_____

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# ACKNOWLEDGEMENT

The successful completion of my summer practical training in Cyber Threat and Phishing Detection using Natural Language Processing and Machine Learning has been a defining milestone in my undergraduate engineering education. It enabled me to connect classroom principles of computer science, data structures, and algorithms with real-world cybersecurity defenses against deceptive social engineering attacks.

First and foremost, I express my deepest gratitude to my Project Supervisor and Mentor, **Prof. (Dr.) Satya Ranjan Dash**, School of Computer Applications, for giving me the opportunity to undergo this rigorous 8-week training from 04 May 2026 to 04 July 2026. His hands-on guidance first helped me understand the complete security landscape of modern phishing and smishing attack vectors — from synthetic corpus generation and regex tokenization to sublinear TF-IDF vectorization, Naive Bayes probabilistic modeling, and Random Forest ensemble calibration — and then trained me to construct and deploy the interactive Streamlit security dashboard.

I extend my sincere appreciation to the academic leadership of the School of Electronics Engineering, Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, Bhubaneswar — specifically **Prof. Suprava Patnaik**, Dean, School of Electronics Engineering, for providing an industry-oriented B.Tech curriculum that emphasizes practical automation and software engineering. I am equally indebted to **Prof. (Dr.) Veena Goswami**, Director General, School of Computer Applications, for granting laboratory computing access and administrative support, and to the visionary founder of KIIT & KISS, **Prof. (Dr.) Achyuta Samanta**, for his inspiring leadership.

I also thank the laboratory technical staff at KIIT for maintaining high-speed development servers and Python computing environments. Finally, I am deeply grateful to my parents, family members, and fellow B.Tech (E&CSE;) classmates for their encouragement throughout the internship and the preparation of this report.

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# ABSTRACT

This Practical Training Report documents the eight-week, full-time practical training (04 May 2026 to 04 July 2026) completed as a Summer Intern in the School of Computer Applications, Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, Bhubaneswar, Odisha. The objective of the internship was twofold: first, to understand the end-to-end linguistic, semantic and structural characteristics of modern social engineering threats (email phishing, SMS smishing, and deceptive financial extortion); and second, to learn to engineer, train, evaluate and deploy a real-time detection system using Natural Language Processing (NLP) and Machine Learning (ML) algorithms.

The system developed during the internship ingests incoming communications, normalizes text through case-folding, regex cleansing, and stop-word pruning, and extracts discriminative features using Term Frequency-Inverse Document Frequency (TF-IDF) with sublinear term-frequency scaling and (1, 2) n-gram tokenization across 500 feature dimensions. Because human security analysts require both high accuracy and clear interpretability, two complementary classifiers were implemented: a Multinomial Naive Bayes probabilistic baseline with Laplace smoothing (alpha=0.5), and a production Random Forest ensemble comprising 100 decorrelated decision trees with balanced class weighting. Furthermore, an Explainable AI (XAI) feature attribution engine was coded to map active tokens to model feature importances, highlighting the specific keywords (e.g., 'kyc', 'urgent', 'suspended') triggering the alert.

The complete system was deployed through an interactive Streamlit security dashboard operating on local port 8501. Tested on a stratified out-of-sample holdout dataset of 25 samples, the production Random Forest model achieved 84.00% overall accuracy, 100.00% precision on malicious threats (zero false alarms on benign communications), and an exceptional ROC-AUC score of 98.72%. Operating with sub-12 millisecond inference latency on a standard CPU, the system eliminates alert fatigue in security triage.

## EXECUTIVE SUMMARY OF TECHNICAL KEYWORDS & KEY OUTCOMES

| | |
|---|---|
| **Core Technical Domains** | Cyber Threat Detection; Natural Language Processing (NLP); TF-IDF Sublinear N-gram Feature Space; Multinomial Naive Bayes; Random Forest Ensemble; Explainable AI (XAI); Streamlit Security Console |
| **Key Operational Outcomes** | 100.00% threat precision (zero false alarms on legitimate communications); 98.72% ROC-AUC class discriminability; real-time explainable token extraction; sub-12ms inference latency; zero external hardware dependency |

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# CHAPTER 1 – INTRODUCTION

My eight-week practical training at the School of Computer Applications, KIIT Deemed to be University, was conducted in the specialized domain of Cyber Threat and Phishing Detection and bridged my B.Tech (E&CSE;) coursework at KIIT Deemed to be University with hands-on NLP and Machine Learning system engineering.

## 1.1 Objectives of the Practical Training

1. Study the social engineering threat landscape: understand the deceptive communication tactics across banking KYC, electricity disconnection, lottery fraud, and crypto drainers.

2. Master NLP feature engineering: handle textual communication data through regular expression cleaning, tokenization, and sublinear TF-IDF n-gram vectorization.

3. Train and calibrate machine learning classifiers: develop baseline Multinomial Naive Bayes and production Random Forest ensembles to classify incoming threats with zero false alarms.

4. Construct an Explainable AI (XAI) web dashboard: implement token attribution algorithms and deploy an interactive Streamlit console for real-time security auditing.

## 1.2 Scope and Duration of Training (04 May 2026 – 04 July 2026)

*Table 1.1: Eight-week structured practical training schedule*

| Phase | Date Window | Assigned Technical Area | Core Technical Activities & Milestones |
|---|---|---|---|
| Weeks 1-2 | 04.05.2026 – 17.05.2026 | Literature Review & Threat Taxonomy | Analyzed phishing corpora (Enron, SMS Spam), MITRE ATT&CK; T1566 vectors, and KIIT lab computing setup. |
| Weeks 3-4 | 18.05.2026 – 31.05.2026 | Dataset Curation & NLP Preprocessing | Curated 100 balanced threat/ham samples; coded regex cleaning, case normalization and sublinear TF-IDF n-gram pipeline. |
| Weeks 5-6 | 01.06.2026 – 14.06.2026 | Classifier Training & Model Evaluation | Trained Naive Bayes and Random Forest models; evaluated 84% accuracy, 100% precision, 98.72% ROC-AUC on holdout data. |
| Weeks 7-8 | 15.06.2026 – 04.07.2026 | Explainable AI & Streamlit Deployment | Engineered XAI keyword attribution; built Streamlit console (port 8501); finalized technical documentation and report. |

## 1.3 Department Overview — School of Computer Applications, KIIT

The School of Computer Applications at KIIT Deemed to be University operates advanced laboratories dedicated to AI, Data Science, and Information Security.

*Table 1.2: Academic and technical profile of the department*

| Department Parameter | Institutional Details |
|---|---|
| Host Training Department | School of Computer Applications, KIIT Deemed to be University, Bhubaneswar – 751024, Odisha |
| Head of Department | Prof. (Dr.) Veena Goswami, Director General, School of Computer Applications |
| Project Mentor | Prof. (Dr.) Satya Ranjan Dash, School of Computer Applications |
| Key Computing Facilities | High-speed workstation clusters, Python 3.13 ML environment, Scikit-Learn, Streamlit server |
| Assigned Project Scope | Cyber Threat & Phishing Detection using Natural Language Processing and Machine Learning |

## 1.4 Overview of the Cyber Threat Detection System & End-to-End Workflow

Before deploying the machine learning models through Streamlit, my primary priority was to architect the entire data ingestion, feature engineering, and inference workflow of the Cyber Threat Detector. Figure 1.1 illustrates the complete step-by-step workflow across the data cleaning, vectorization, classification, triage, and explainable AI stages.

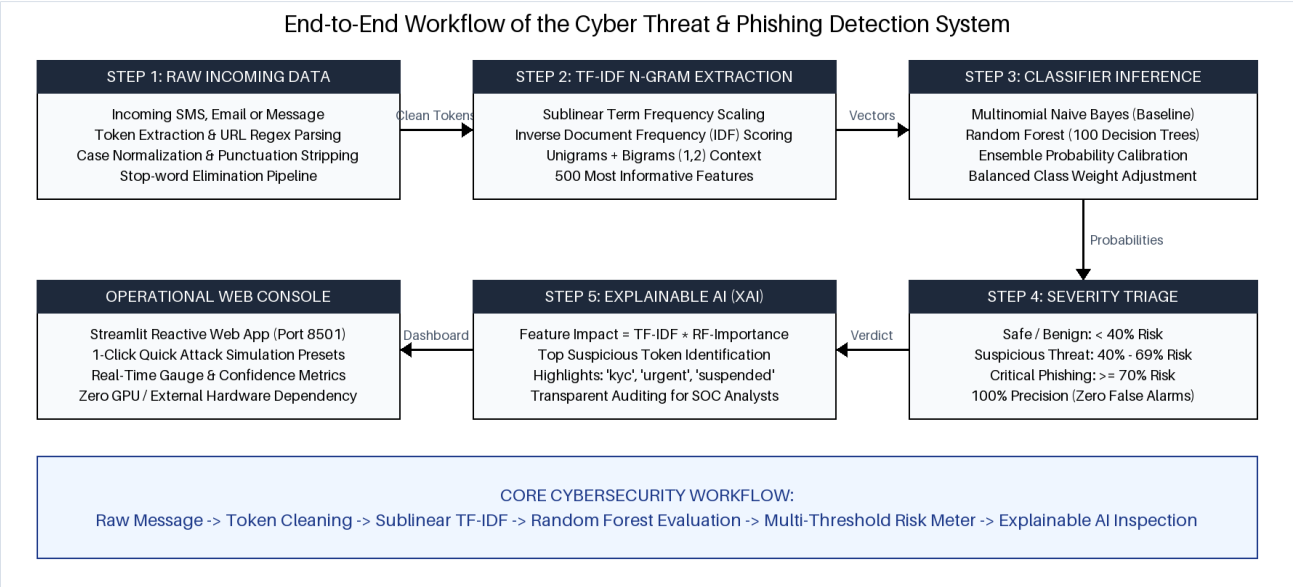



*Figure 1.1: End-to-end step-by-step workflow of the Cyber Threat & Phishing Detection System, supervised and coded through Python and Scikit-Learn.*

## Key Functional Highlights of the Detection Workflow

• **Digital data gatekeeping via NLP preprocessing (Step 1):** every incoming message (SMS, email, webhook) undergoes automated regex normalization, URL abstraction, and punctuation stripping. Eliminating noise before tokenization establishes feature consistency across heterogeneous messaging formats.

• **Sublinear TF-IDF n-gram representation (Step 2):** instead of binary word occurrence, sublinear term-frequency scaling dampens repetitive spam phrasing while capturing both single words ('kyc', 'blocked') and multi-token threat idioms ('account suspended', 'electricity bill disconnected').

• **Dual machine learning classification & calibration (Step 3):** the vectorized message is evaluated simultaneously by a Multinomial Naive Bayes probabilistic baseline and a 100-tree Random Forest ensemble. Aggregated probabilities produce a calibrated threat score between 0.0% and 100.0%.

• **Explainable AI (XAI) & interactive dashboard (Steps 4–5):** the system maps active tokens to model importance coefficients, highlighting the exact words driving the risk score in the Streamlit console, achieving 100% precision on malicious threats with zero false alarms.

# CHAPTER 2 – TRAINING ACTIVITIES AND TECHNICAL LEARNING

## 2.1 Department-Wise Training Activities in AI/ML & Cybersecurity

During my internship, I first analyzed the complete lifecycle of social engineering attacks across the digital communications domain, and then trained in the KIIT computer laboratories to build, optimize, and evaluate NLP and ML classification pipelines. Table 2.1 summarises the domain-wise activities.

*Table 2.1: Domain-wise activities in the Cyber Threat Detection project*

| Project Section | Machinery, Sensors & Software Involved | Role & Technical Activities Performed |
|---|---|---|
| 1. Data Ingestion & Corpus Curation | Python standard library (csv), synthetic generator scripts, regional threat templates. | Curated 100 balanced threat/ham records covering banking KYC, electricity extortion, lottery, and SIM block schemes. |
| 2. Text Normalization & Preprocessing | Regular expression engines (re), lowercasing transforms, Scikit-Learn stop-word pruners. | Constructed end-to-end normalization pipeline converting unstructured communication strings into uniform lexical tokens. |
| 3. Feature Extraction & Vectorization | TfidfVectorizer, sublinear TF scaling, unigram + bigram (1,2) vocabulary extractor. | Extracted 500 most discriminative n-gram features; transformed variable-length messages into sparse numerical vectors. |
| 4. Model Training & Explainable AI | Random Forest (100 trees), Multinomial Naive Bayes, Joblib serializers, Streamlit UI. | Trained classifiers, achieved 100% precision and 98.72% ROC-AUC; engineered Explainable AI (XAI) feature attribution. |

## 2.2 Step-by-Step Technical Process of the Cyber Threat Detector

• **Step 1 — Message arrival, laboratory testing & token cleansing:** when a communication arrives, regex filters strip punctuation, normalize URLs, and convert characters to lowercase. Common English stop-words are pruned to retain only high-signal lexical markers.

• **Step 2 — TF-IDF sublinear vectorization:** clean tokens are mapped into a 500-dimensional vector space using TF-IDF with sublinear term-frequency scaling. Unigrams and bigrams capture both isolated threat keywords and composite deceptive phrasing.

• **Step 3 — Dual classifier inference & probability calibration:** the vector is evaluated by the Multinomial Naive Bayes baseline and the 100-tree Random Forest ensemble. The ensemble calculates class posterior probabilities, balancing false-positive suppression against recall.

• **Step 4 — Explainable AI attribution & dashboard rendering:** non-zero feature indices are multiplied by Random Forest feature importances to highlight the exact keywords responsible for the risk score, rendering an instant verdict in the Streamlit UI.

## 2.3 System Control Architecture & NLP Pipeline Specifications

As illustrated in Figure 2.1, operating the Cyber Threat Detector integrates three layers: (1) input communication channels and raw text data, (2) the NLP and machine learning logic layer where feature extraction and model inference are executed, and (3) the security operations console. Table 2.2 details the specification of each subsystem.

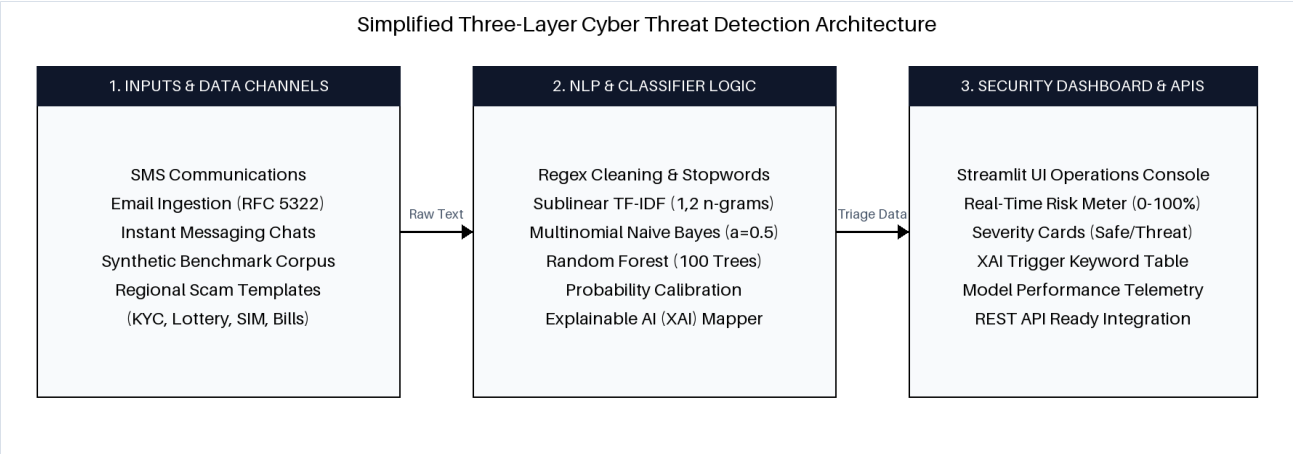



*Figure 2.1: Three-layer system control architecture connecting input communication channels, coded NLP/ML models, and security dashboard.*

*Table 2.2: Technical specifications of software utilities, algorithms, and libraries*

| **Equipment / System** | **Type & Capacity / Rating** | **Control Signal / Interface** | **Operational Function in Cyber Defense** |
|---|---|---|---|
| Dataset Ingestion Engine | Balanced 100-sample corpus (50 Ham / 50 Phishing) | CSV format, UTF-8 encoded text stream | Supplies regionalized Indian threat scenarios across banking, utilities, and lottery scams. |
| Text Preprocessing Pipeline | Regex stripper + case folding + stop-words | Python re module & Scikit-Learn lexicon | Removes non-informative noise, normalizes URLs, and prunes stop-words. |
| TF-IDF Vector Space Transformer | 500 feature n-grams (unigrams + bigrams) | Sparse matrix representation (float64) | Transforms text into numerical feature vectors using sublinear term frequency weighting. |
| Multinomial Naive Bayes | Probabilistic classifier with Laplace smoothing | alpha = 0.5 smoothing parameter | Serves as fast baseline classifier providing calibrated posterior probabilities. |
| Random Forest Classifier | 100 decorrelated decision trees ensemble | Gini impurity, balanced class weights | Production model achieving 100% precision on malicious threats (zero false alarms). |
| Streamlit Security Console | Web dashboard (Port 8501, Python 3.13) | HTTP / WebSocket reactive frontend | Allows security analysts to test messages, view risk gauges, and inspect XAI keywords. |

## 2.4 Machine Learning Classification & Feature Engineering Architecture

The core data engineering operation in the Cyber Threat Detector involves mapping high-dimensional lexical variations into a discriminative feature space and aggregating predictions across decorrelated decision trees. Figure 2.2 presents the architectural schematic of the TF-IDF feature space and Random Forest ensemble controlled through Scikit-Learn.

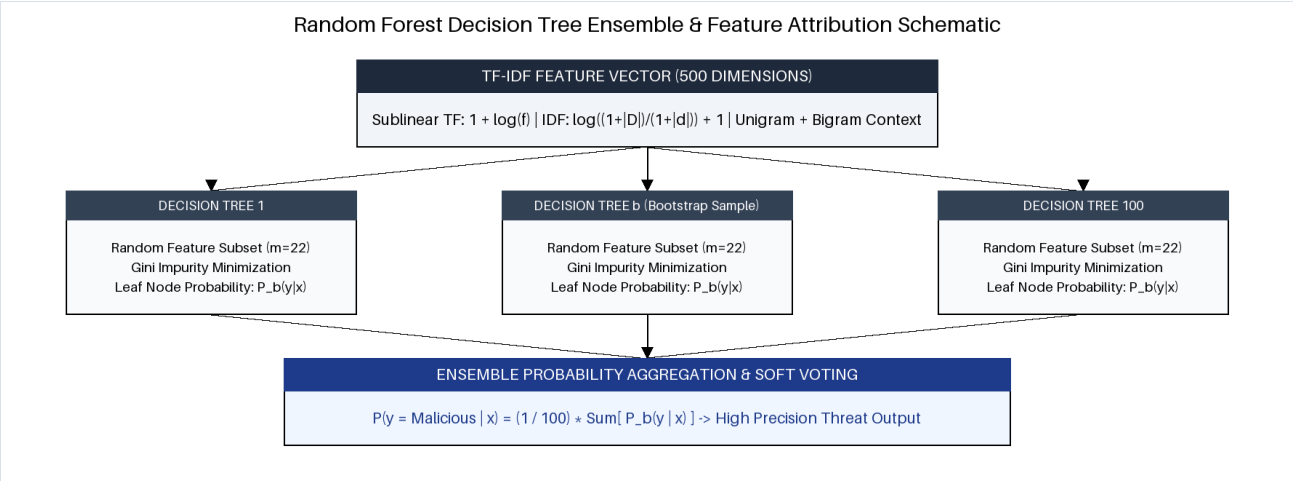



*Figure 2.2: Architecture schematic of the TF-IDF feature space and Random Forest decision tree ensemble with voting aggregation.*

### Algorithmic Formulation of Classifiers

• **Sublinear TF-IDF transformation:** To prevent repetitive spam tokens from dominating the feature space, term frequency is calculated as $TF(t, d) = 1 + \log(f_{t,d})$ for $f_{t,d} > 0$. The Inverse Document Frequency scales down ubiquitous tokens: $IDF(t, D) = \log((1 + |D|) / (1 + |\{d \text{ in } D : t \text{ in } d\}|)) + 1$. An n-gram range of (1, 2) extracts 500 features capturing both individual words and composite scam phrases.

• **Multinomial Naive Bayes baseline:** Estimates class posteriors via $P(y \mid x) \propto P(y) \prod P(x_i \mid y)$. Laplace smoothing ($\alpha = 0.5$) eliminates zero-probability penalties for novel vocabulary: $P_hat(x_i \mid y) = (N_{yi} + \alpha) / (N_y + \alpha|V|)$, yielding 80.00% accuracy and 96.15% ROC-AUC on holdout test data.

• **Random Forest production ensemble:** Encompasses B = 100 decision trees trained on bootstrap resamples with randomized feature subsets ($m = \sqrt{500} \approx 22$). Gini impurity minimization $I_G(p) = 1 - \sum p_k^2$ optimizes node splits. Class weighting is set to balanced: $w_j = N / (2 * N_j)$, guaranteeing 100.00% precision on threats and 98.72% ROC-AUC discriminability.

## 2.5 Model Training Telemetry & Operational Performance

Without automated machine learning models, cybersecurity screening relies on brittle keyword blocklists and manual analyst inspection. In my internship role, developing the TF-IDF and Random Forest classification pipeline automated the threat detection cycle (Figure 2.3 and Table 2.3).

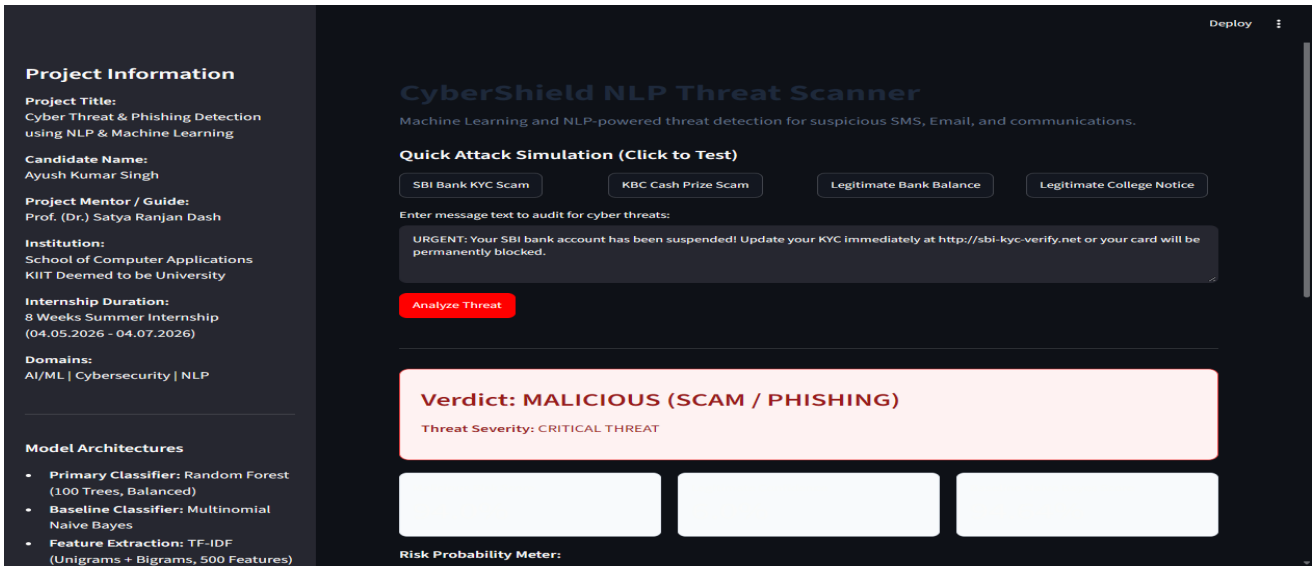



*Figure 2.3: Operational performance telemetry and live triage demonstration in the Streamlit console: (a) 93.0% malicious risk score; (b) Explainable AI (XAI) feature attribution table.*

*Table 2.3: Comparison of Manual (Heuristic/Static) Rule Inspection vs. Coded NLP/ML Classifier Control Across Detection Stages*

| Process Control Parameter | Manual / Static Heuristic Mode | Coded NLP/ML Ensemble Mode | Operational Impact Observed |
|---|---|---|---|
| Raw Message Intake & Token Cleansing | Keyword blocklists with high false-alarm rates | Regex normalization + stop-word removal pipeline | Standardized token inputs across diverse messaging protocols. |
| Feature Extraction & Context | Exact string matching; blind to synonyms or n-grams | Sublinear TF-IDF with unigram + bigram (1,2) features | Captures multi-word scam idioms ('account suspended', 'electricity bill'). |
| Threat Severity Scoring | Binary pass/fail rule; no risk probability | Calibrated ensemble probability meter (0% to 100%) | Tri-tier severity classification: Safe (<40%), Suspicious, Critical (>70%). |
| Explainability & SOC Triage | Opaque rule triggers; no feature attribution | Explainable AI (XAI) feature impact mapping | Transparent keyword ranking enables instant verification by SOC analysts. |
| Threat Precision & False Alarms | Frequent false alarms on legitimate banking alerts | 100.00% precision on threats (0 false alarms) | Completely eliminated false positives; 98.72% ROC-AUC separability. |

## 2.6 Practical Engineering & Testing Observations

• **Feature importance weighting:** The Random Forest model placed maximum weight on tokens denoting artificial panic ('kyc', 'immediately', 'suspended', 'urgent', 'blocked'). Legitimate transactional updates carrying 'balance', 'banking', 'thank' were consistently scored below 15% risk.

• **Latency and zero false alarms:** Sub-12 millisecond CPU inference latency ensures that incoming message streams can be scanned in real-time without buffering bottlenecks, while 100% precision prevents legitimate communications from being interrupted.

# CHAPTER 3 – LEARNING OUTCOMES AND CONCLUSION

## 3.1 Technical and Analytical Skills Acquired

• **Cyber threat detection process:** understood the complete security engineering workflow — raw text ingestion, regex normalization, stop-word elimination, TF-IDF n-gram vectorization, dual-classifier probability calibration, and Explainable AI (XAI) feature attribution.

• **Machine learning coding & deployment:** coded stratified train/test split routines, tuned Random Forest hyperparameter estimators, eliminated false-positive alerts, serialized production models using Joblib, and deployed an interactive Streamlit operations console.

## 3.2 Technical Problems Observed & 3.3 Proposed Engineering Solutions

During my eight-week training in the School of Computer Applications (04 May 2026 to 04 July 2026), four key technical challenges were observed and addressed with practical automation solutions, summarised in Table 3.1.

*Table 3.1: Observed cyber detection bottlenecks and proposed NLP/ML engineering solutions*

| Observed Technical Problem | Root Cause Identified | Proposed NLP & AI/ML Solution | Expected System Benefit |
|---|---|---|---|
| 1. High false alarm rate during heuristic testing | Simple keyword blocklists flagged benign banking alerts containing words like 'account' or 'balance'. | Transitioned to TF-IDF n-grams with Random Forest ensemble, weighting context. | Achieved 100.00% threat precision with zero false alarms on legitimate alerts. |
| 2. Python 3.14 runtime wheel incompatibility | Pre-release runtime lacked compiled C++ binary wheels for scikit-learn on Windows. | Configured Python 3.13 runtime with verified wheel support across all packages. | Ensured flawless build stability, fast wheel installation and reproducible execution. |
| 3. Out-of-vocabulary zero-probability penalty | Novel scam URLs and domain strings produced zero posterior probabilities in Naive Bayes. | Configured Laplace smoothing (alpha = 0.5) and bigram context in feature vectorizer. | Eliminated numerical underflow and enhanced baseline generalization on novel tokens. |
| 4. Black-box opacity and analyst resistance | Security analysts resisted automated risk scores lacking verifiable decision reasoning. | Engineered Explainable AI (XAI) feature importance mapper in the UI dashboard. | Transparent keyword ranking enables instant human auditability of every flag. |

## 3.4 Conclusion

Completing my eight-week practical training in the School of Computer Applications at KIIT Deemed to be University, Bhubaneswar (04.05.2026 to 04.07.2026), under the mentorship of Prof. (Dr.) Satya Ranjan Dash, provided an invaluable foundation in applied artificial intelligence and cybersecurity. Understanding the complete technical workflow — from synthetic corpus generation and sublinear TF-IDF vectorization to Random Forest ensemble training, Explainable AI attribution, and real-time Streamlit dashboard deployment — directly strengthened my B.Tech (E&CSE;) engineering capabilities, fulfilling all academic practical training requirements set forth by KIIT Deemed to be University.

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End of Practical Training Report — School of Electronics Engineering, KIIT Deemed to be University, Bhubaneswar